Review plan · for the teacher

Sensors: unit check

IoT · Level 1 · Unit check

Time	Materials
45 min	Board, chalk, paper. No devices needed.

Learning goals

- What is a sensor?
- Reading numbers: units and scales
- Build a rain gauge
- Data in class: record, chart, decide

Activity: Team quiz

1. Make teams of four
2. Read each question aloud; teams agree on one answer
3. Reveal the answer and ask one team to explain why
4. Write the hardest question on the board for tomorrow

Discussion

- Which idea will you use first?
- Which question surprised you, and why?
- What will you tell your family about this unit?

Exit ticket

- One idea I will use, and when I will use it

Try it in class tomorrow

Pick the habit from this unit your pupils need most, teach it this week, and tell a colleague how it went.

Quiz sheet · for each team

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Team _____ Date _____

1 **Which of these is a sensor?**

- ◯ A scoreboard showing the match score
- ◯ The part of a solar lamp that notices daylight fading
- ◯ A pump that waters the school garden

2 **Match each part to its job.**

Sensor	• • • •	Acts, such as pumping water or sounding a buzzer
Display	• • • •	Notices something and turns it into a number
Actuator	• • • •	Shows the reading to people

3 **"28" is a complete temperature reading.**

- ◯ True
- ◯ False

4 **Your class compares Monday's and Tuesday's temperatures. Which make the comparison fair?**

- ☐ Both read in the same shady spot
- ☐ One in °C and one in °F
- ☐ Both read at 8 am
- ☐ One read in the sun, one in the shade
- ☐ Both read with eyes level with the liquid

5 **Put the rain gauge steps in order.**

- ☐ Place it in open ground
- ☐ Add a mm scale and the funnel
- ☐ Read it at the same time each day
- ☐ An adult cuts off the bottle's top
- ☐ Add stones and water; mark the 0 line

6 **Why must a bottle rain gauge have straight sides?**

- ◯ So every millimetre on the scale matches a millimetre of rain
- ◯ So it holds more water
- ◯ So it is easier to carry

7 **Complete the sentence about using the rain gauge.**

Put the gauge in _____ ground, read it at the same _____ each day, then pour it back to _____.

Word bank: time · shady · open · zero

8 **A connected garden system does the same steps as your class. Which statements are true?**

- ☐ It works without power or a network
- ☐ A sensor can record readings automatically
- ☐ It never needs checking
- ☐ A rule can switch on a pump when the soil is dry

9 **A blank box in the rain table means no rain fell that day.**

- ◯ True
- ◯ False

Answers · for the teacher

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1 **The part of a solar lamp that notices daylight fading**

The light sensor notices the daylight fading and turns that into a signal that switches the lamp on.

2 **Sensor — Notices something and turns it into a number**
Display — Shows the reading to people
Actuator — Acts, such as pumping water or sounding a buzzer

Sense, show, act. In between, a person or a small computer decides what to do with the reading.

3 **False**

"28" alone needs a unit (°C), plus a time and a place, so others can check it and compare it fairly.

4 **Both read at 8 am; Both read in the same shady spot; Both read with eyes level with the liquid**

Reading both at 8 am removes the morning-to-afternoon change, the same shady spot matters because the sun heats the thermometer itself, and reading both with eyes level stops the angle shifting the number.

5 **Top to bottom: 4, 3, 5, 1, 2**

Build it, place it, then read it at the same time daily — and pour it back to 0 after each reading.

6 **So every millimetre on the scale matches a millimetre of rain**

Straight sides keep every millimetre on the scale equal to a millimetre of rain.

7 **Put the gauge in open ground, read it at the same time each day, then pour it back to zero.**

Open ground keeps roofs and trees from changing the catch, and a shady spot is usually under one of them.

8 **A sensor can record readings automatically; A rule can switch on a pump when the soil is dry**

A sensor can record a reading every few minutes and send it over a network, and a rule that switches on a pump when the soil is dry is the decide step, done automatically.

9 **False**

A blank could mean no rain or no reading.